



Service Pack Guideline

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1. Purpose

This guideline provides a consistent framework for planning, developing, verifying, validating and releasing Service Packs (SPs) for released DXR products. It supports software security, stability, and compliance through a controlled 90-day release cadence.

2. Scope

Organization name: DXR QMS – 0015

This process applies to released DXR products for which third-party support remains available for the applicable components. It covers security updates for non-safety, non-regulatory, and non-privacy fixes distributed through Service Packs.

Obsolescence related changes and firmware updates are excluded and shall follow the established design change and release processes.

3. References

Sl.No	Document Number	Document Title
1	2009000799	Defect Management Operating Procedure
2	2009000873	4.1.1 Modify Launched Design Procedure
3	2009001084	Product and Software Identification Standard

4. Abbreviations

Abbreviation	Definition
AWS PC	Applicable workstation or PC configuration used for the product
CQ	ClearQuest
DXR	Diagnostic X-Ray
EOL	End of Life
MLD	Modify Launched Design
QMS	Quality Management System
RFD	Release for Delivery
SP	Service Pack
V&V	Verification and Validation

5. Roles and Responsibilities

Role	Responsibilities
MLD Program Manager	• Initiates the Service Pack project. • Coordinates cross-functional inputs. • Defines and controls scope. • Creates required defect and security-patch records. • Manages the 90-day schedule. • Coordinates verification. • Ensures timely release in In-center • Communicates the release status to relevant stakeholders.
Cross-Functional Team	• Identifies and reports defects. • Provides scope inputs. • Supports impact assessment, verification. • Ensures required documentation and standards compliance. • Reviews or approves release records within its area of responsibility.
Quality / Regulatory	• Reviews safety and regulatory impact. • Confirms the adequacy of compliance evidence. • Provides approval when required by the applicable procedure.

6. Guiding Principles

- Service Packs are cumulative and are planned at 90-day intervals for each supported product software version in Installed base.
- The quarterly cadence is intended to support applicable regulatory requirements, security-policy obligations and customer commitments.
- Each Service Pack shall be managed as a controlled project under the Modify Launched Design process [2]
- Package versioning shall follow the Product and Software Identification Standard [3]
- A monthly SteerCo meeting shall be conducted as part of the governance model to review Service Pack progress and release status against the planned cadence.

7. Service Pack Decision Flow

The following decision flow determines whether the proposed scope is eligible for inclusion in a Service Pack release.

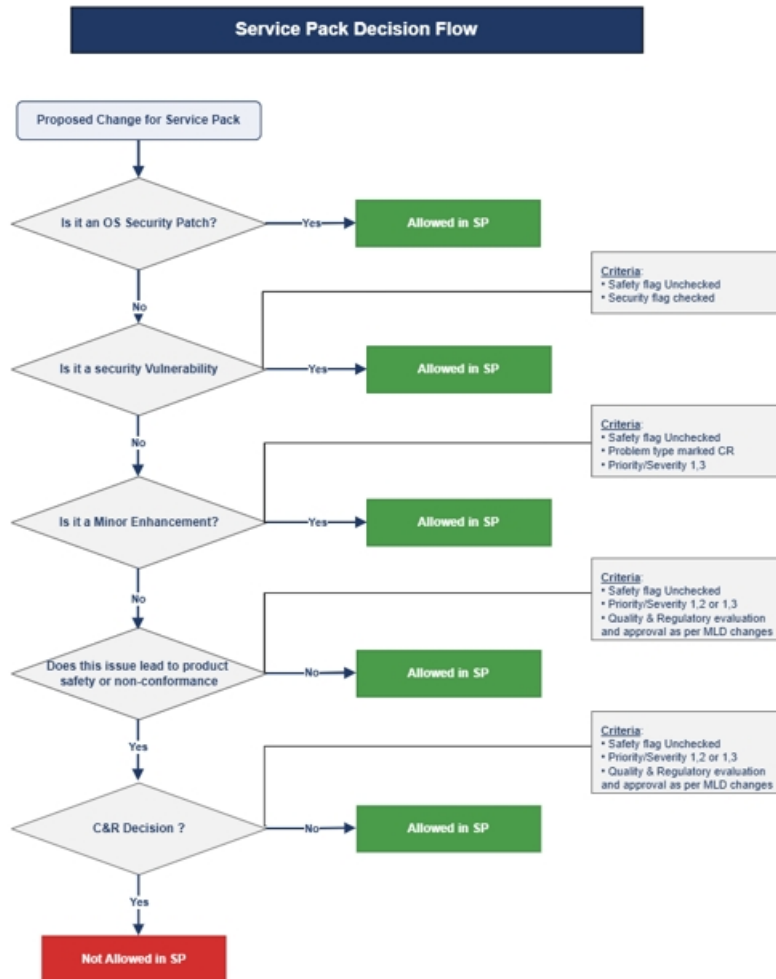
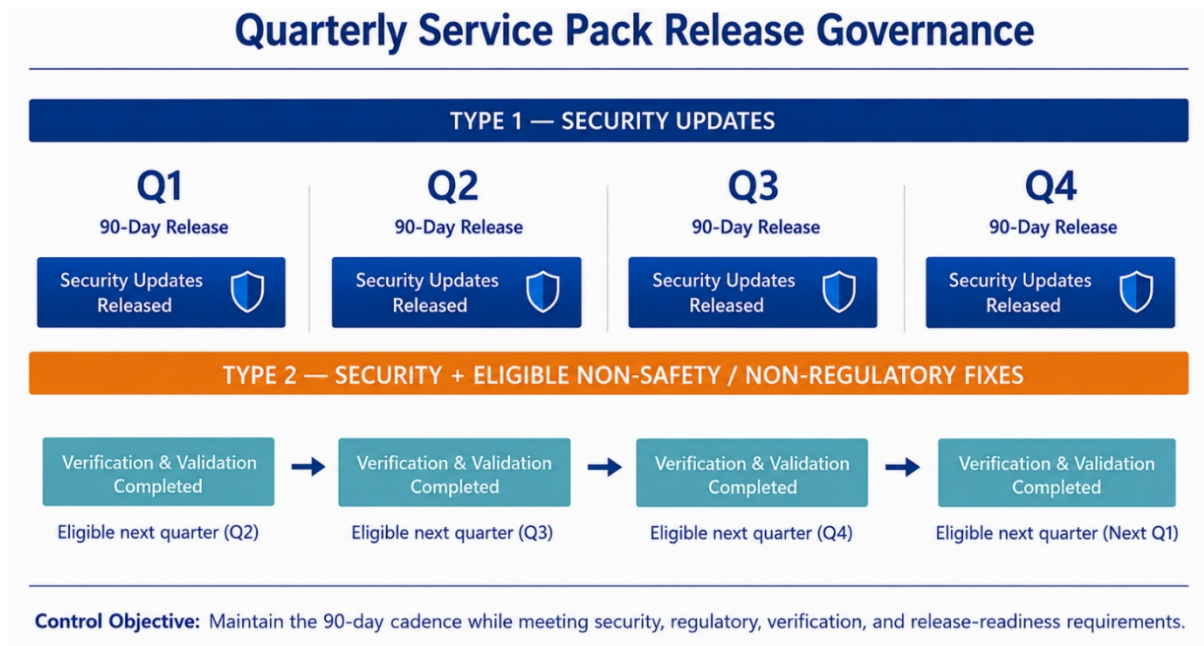


Figure 1. Decision flow for determining whether a proposed scope is eligible for inclusion in a Service Pack release.

8. Service Pack Classification

In-Order to maintain the 90-day release cadence, Service Packs shall be classified as follows.

Type	Allowed Scope	Release Expectation
Type 1 Security Updates	Updates required to remediate identified security vulnerabilities and maintain compliance with applicable security requirements.	Planned for each quarterly release.
Type 2 Security and applicable Fixes	Required security updates together with fixes that do not affect safety, regulatory compliance, or privacy.	Planned fixes shall complete all required quality checks at least one quarter prior to the intended release.



9. Verification and Validation guidance

9.1 Type 1 Service Packs

Where security remediation is limited to software changes and does not affect intended use, safety, performance, privacy, or overall system functionality. The verification shall be limited to the applicable AWS PC configurations. The selected scope shall be supported by a documented impact assessment and approved in accordance with established procedures.

9.2 Type 2 Service Packs

Based on the impact assessment, verification shall be conducted either at the system level or exclusively on the applicable AWS PC configurations. Test-system selection should cover impacted product types and currently supported versions, with at least one representative version of each impacted product type. Applicable fixes should complete verification at least one quarter before the intended release.

10. Customer Installation Requirements

To support customer-installable software capabilities and help ensure a high self installation success rate, the following measures shall be considered.

- Any changes requiring backup and restore shall not be included in Servicepack.
- Only the predefined installation checks shall be used. In case of additional checks or tests needed, the change shall be reviewed with the Service stakeholders before inclusion.

11. Exceptions and Exclusions

- Hardware obsolescence-related changes and firmware updates should not be included in Service Packs because they may affect product design, require extensive verification, and jeopardize the quarterly cadence.
- When third-party support for an applicable component reaches end of life, a Service Pack should be considered only for a specifically requested immediate fix whose criticality is documented and approved project team.

- Non-cumulative Service Packs shall not be issued, to ensure controlled release management, configuration consistency, and compliance with version-control requirements.

- Document references

Document Number	Document name
2009000799	Defect Management Operating Procedure
2009000873	4.1.1 Modify launched design Procedure
2009001084	Product and Software Identification Standard

Revision History

Version	Document Action	Change Plan ID	Approver(s)	Content Reviewer(s)	Compliance Reviewer(s)	Document Change Description
1	Initial version	CP_QMS-0015_000176	Peter, Vidya Cen (310157795) at 12-Aug-2025 11:39 CEST	Verdonck, Rene (nly14328) at 31-Jul-2025 14:14 CEST Asutkar, Kanchan (320103582) at 06-Aug-2025 07:04 CEST JANGIR, Pramod (320245826) at 31-Jul-2025 15:23 CEST	N/A	Initial release
2	Content update	CP_QMS-0015_000321	Peter, Vidya Cen (310157795) at 17-Sep-2026 11:27 CEST	Jain, Namrata Ankush (320071878) at 17-Sep-2026 08:27 CEST Cassens, Isabel (310076060) at 11-Sep-2026 10:59 CEST Koul, Swati (310281467) at 11-Sep-2026 09:44 CEST Verdonck, Rene (nly14328) at 17-Sep-2026 10:58 CEST	N/A	Key updates include: Emphasis on a 90-day Service Pack release cadence Obsolescence-related changes are out of scope Non-cumulative Service Packs are out of scope Decision tree added to assess and scope CQs in Service Packs Governance mechanism added to monitor and ensure adherence to the 90-day release cadence

Publication Signing Manifest (Current Version)

Change Review Team Representative	Approval Meaning
Cassens, Isabel (310076060) at 17-Sep-2026 11:35 CEST	Approve Change Plan as 'Ready for Use'

Status	Effective Date
Effective	21-Sep-2026

Notes: Approvals are given by electronic signature.
All time stamps are CET time zone in 24-hour format.

Revision History and Publication Signing Manifest System Template

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